3.3 Proofs with Parallel Lines

Name _____ Period ____ Date _____

Geometry 3.3 · Student sheet

I can…

- use the Corresponding Angles Converse to decide whether lines are parallel.
- use the other converses to decide whether lines are parallel.
- use the Transitive Property of Parallel Lines.

Vocabulary · fill in the missing word

Converse — the statement formed by switching the _____ and the conclusion

Transitive Property of Parallel Lines — _____

Part 1 · Discovery: make them parallel

Draw a transversal. At two points on it, draw lines that make 60° corresponding angles. Measure the distance between the lines in two places.

Distance at the left	Distance at the right	Are the lines parallel?

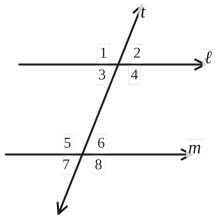
If corresponding angles are congruent, then the lines are _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Corresponding Angles Converse: if two lines are cut by a transversal so that corresponding angles are congruent, then the lines are _____.

Example 1

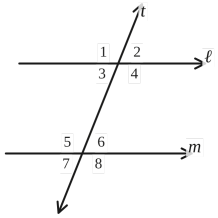
Find the value of x that makes $\ell \parallel m$: $m\angle 1 = (3x + 5)°$ and $m\angle 5 = 65°$.



Other converses: if alternate interior angles or alternate exterior angles are congruent, the lines are parallel; if consecutive interior angles are _____, the lines are parallel.

Example 2

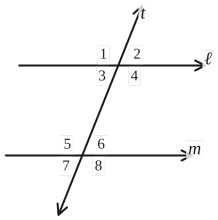
$m\angle 4 = 75°$ and $m\angle 6 = 105°$. Are ℓ and m parallel? Explain.



Example 3

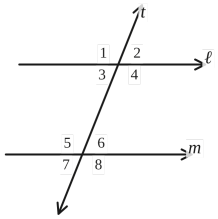
a. $m\angle 2 = 80°$ and $m\angle 7 = 80°$. Is $\ell \parallel m$?

b. $m\angle 3 = 70°$ and $m\angle 5 = 70°$. Is $\ell \parallel m$?



Example 4

Given: $\angle 1 \cong \angle 8$. Prove: $\ell \parallel m$.



Transitive Property of Parallel Lines: if two lines are parallel to the same line, then they are _____ to each other.

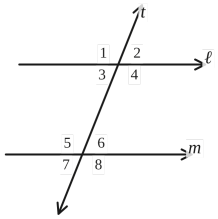
Example 5

Each rung of a ladder is parallel to the top rung. Are the rungs parallel to each other? Explain.

Part 3 · Partner practice

Partner 1

Find x so that $\ell \parallel m$: $m\angle 3 = (2x + 18)°$ and $m\angle 6 = (4x - 12)°$.



Partner 2

$m\angle 4 = 112°$ and $m\angle 6 = 68°$. Are ℓ and m parallel? Explain.

